

The background of the slide features a complex, abstract pattern. It consists of numerous concentric circles and arcs, each composed of small, rectangular segments in various shades of gray. These segments are arranged in a way that creates a sense of depth and movement, as if they are floating or orbiting. The overall effect is a dense, textured field of geometric shapes that fills the entire frame.

BAM5103-01

PROGRAMMING FOR ANALYTICS

GSB FALL 2026

INSTRUCTOR: PIM KAMPHUIS

ADAPTED FROM COURSE MATERIALS BY DR. YUAN TIEN

TODAY'S AGENDA

-
- Course introduction
 - Discovering the toolkit:
 - Visual Studio Code
 - UV
 - GitHub

COURSE INFORMATION

Instructor

- Pim Kamphuis
 - Preferred contact: email p.i.m@skku.edu
 - Office hours: After class or by appointment

Course Schedule

- Saturdays 13:30 – 17:00 [Classroom 9B114]
 - No class on Sept 26th
 - Inform me in advance if you cannot attend the course

Teaching Assistant

- Yoorim Shim
 - E-mail: yoorim7536@skku.edu
 - Preferred contact: iCampus / Canvas message

	SKK GSB faculty4	900025	900025	Programming for Analytics_BAM5103_01	TA
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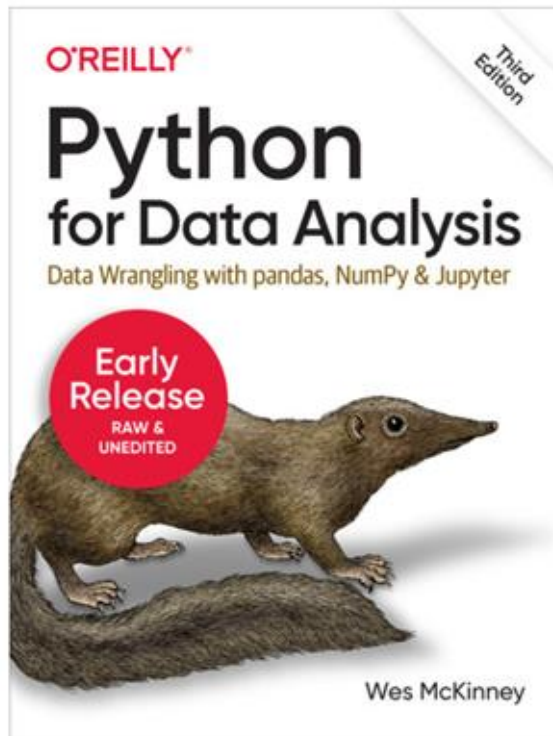
TA Sessions Schedule

- TBD: 7:30pm on Wednesdays or Thursdays

COURSE OUTLINE

- Session 1: The Modern Python Toolkit (VS Code, UV, GitHub)
- Session 2: Python – recap and extension
- Session 3: Pandas – recap and extension
- Session 4: Data transformations, sampling, plotting
- Session 5: Combining and merging data sets, GitHub Copilot
- Session 6: Review and final exam

REFERENCE MATERIALS



- Mostly provided via Google Drive and iCampus.
- Reference Book “Python for Data Analysis”
 - Sign in with SKKU personal mail account:
 - e-Book | Sungkyunkwan University Academic Information Library (skku.edu)
 - Look for O'Reilly Higher Education and click to get access
 - Enter your @skku.edu account (NOT @g.skku.edu) to access it.
 - Access “Python for Data analysis” via the author’s website:
 - <https://wesmckinney.com/book/>

HOUSEKEEPING

- Check in to record your attendance
- Bring your **laptop**, charged if possible
- Questions? Email me at p.i.m@skku.edu

GRADING / FINAL EXAM

- Class participation (10%)
 - Tap to record in-person attendance
- Weekly hands-on assignments and quizzes (50%)
 - Designed to encourage class engagement and practice
 - Late penalty will be 15%
- Final Exam (40%) - October 10, 2026
 - **In-person exam** on campus
 - 70 minutes / Multiple Choice and Writing questions
 - Open-book, no electronic devices allowed

ATTENDANCE

- In accordance with the SKK GSB attendance guideline, students who are absent from class **more than ¼ of required classes without appropriate justification** (excused absence) or students who are absent **more than ½ of required classes** in all circumstances, shall receive a **grade of F** for the course.
- It is the **responsibility of students** to record their attendance properly through SKKU's attendance system (attend.skku.edu)

QUESTIONS?

Reach out to
Yoorim

or

email me at

p.i.m@skku.edu



THE MODERN PYTHON TOOLKIT

IDE, UV AND GIT



TODAY WE WILL INSTALL AND DISCOVER

- Visual Studio Code – Open Source Code Editor
- UV – Ultra Fast Python Project Manager
- Git and GitHub – Version control and codebase management

COLAB VS VISUAL STUDIO CODE

Jupyter Notebooks in **Google Colab**

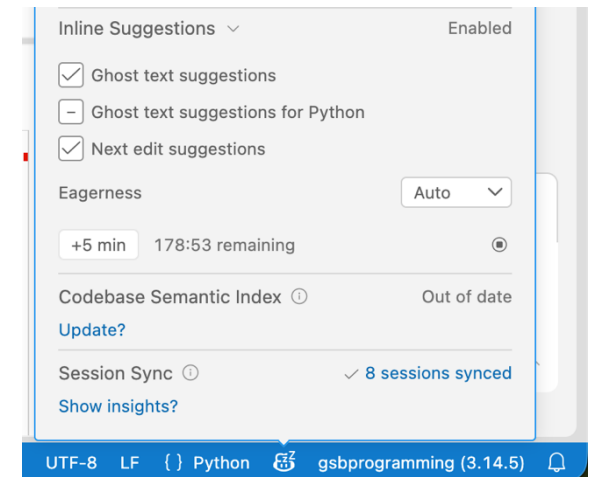
- Easy set-up (nothing to install)
- Use Google's compute (GPU)
- Sync with Google Drive
- Need internet
- Locked into Google's cloud (drive, gemini)
- Python environment: reset every session
- Limited Git, Extensions and terminal
- Less suitable for automation and 'production' pipelines

Jupyter Notebooks in **VS Code**

- Requires some setup to do
- Performance depends on your computer's hardware
- Local storage (git 💡)
- No internet needed
- No vendor lock-in; use extensions and your AI agent of choice
- Python environment: fully managed state
- Excellent integration with Git, Extensions and terminal

SET-UP OUR TOOLKIT

- Installation guide: <https://camphouse.me/posts/get-your-systems-ready/>
 - Check or Install UV: <https://docs.astral.sh/uv/getting-started/installation/> (or run command `uv --version`)
 - Check or Install: <https://git-scm.com/install/> (or run command `git --version`)
 - Check or Install Visual Studio Code: <https://code.visualstudio.com/download>
 - Check or Install VS extensions



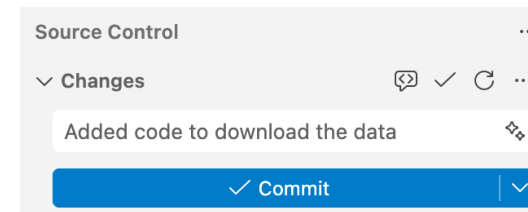
CREATE A PYTHON PROJECT

- Create a folder for our class project (e.g. BAM5103)
- Open the folder in Visual Studio Code
- Create a Python project (environment) in the folder
 - Run `uv init --no-package && uv venv` in the terminal
- Explore the files created
- Discover / recall common Python packages: [[GeeksforGeeks](#)]
- Create a new file: `download.py`
- Let's create a data import pipeline

```
import requests
```

```
r = requests.get("https://github.com/pimatskku/sturdy-  
memory/raw/refs/heads/main/dataset.zip")  
open("dataset.zip", "wb").write(r.content)
```

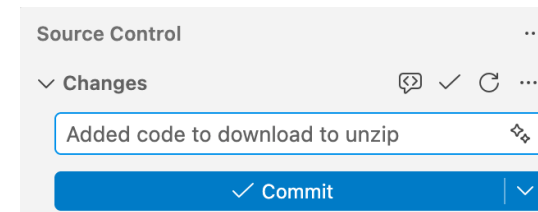
It works? Save and Commit!



```
import zipfile
```

```
with zipfile.ZipFile("dataset.zip", "r") as zip_ref:  
    zip_ref.extractall("data")
```

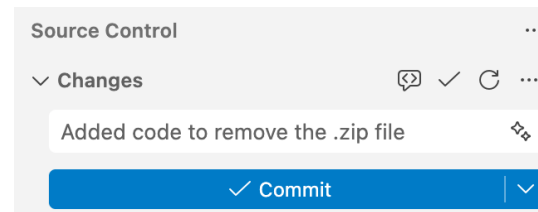
It works? Save and Commit!




```
import os
```

```
os.remove("dataset.zip")
```

It works? Save and Commit!





Data Science in VS Code tutorial

<https://code.visualstudio.com/docs/datascience/data-science-tutorial>



GITHUB

- Create a GitHub account if you don't have one yet: <https://github.com/signup>
- Exercise 1: <https://github.com/skills/introduction-to-github>
- Exercise 2: <https://github.com/skills/communicate-using-markdown>